JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO P.S.AGENCIES IOC DEALERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

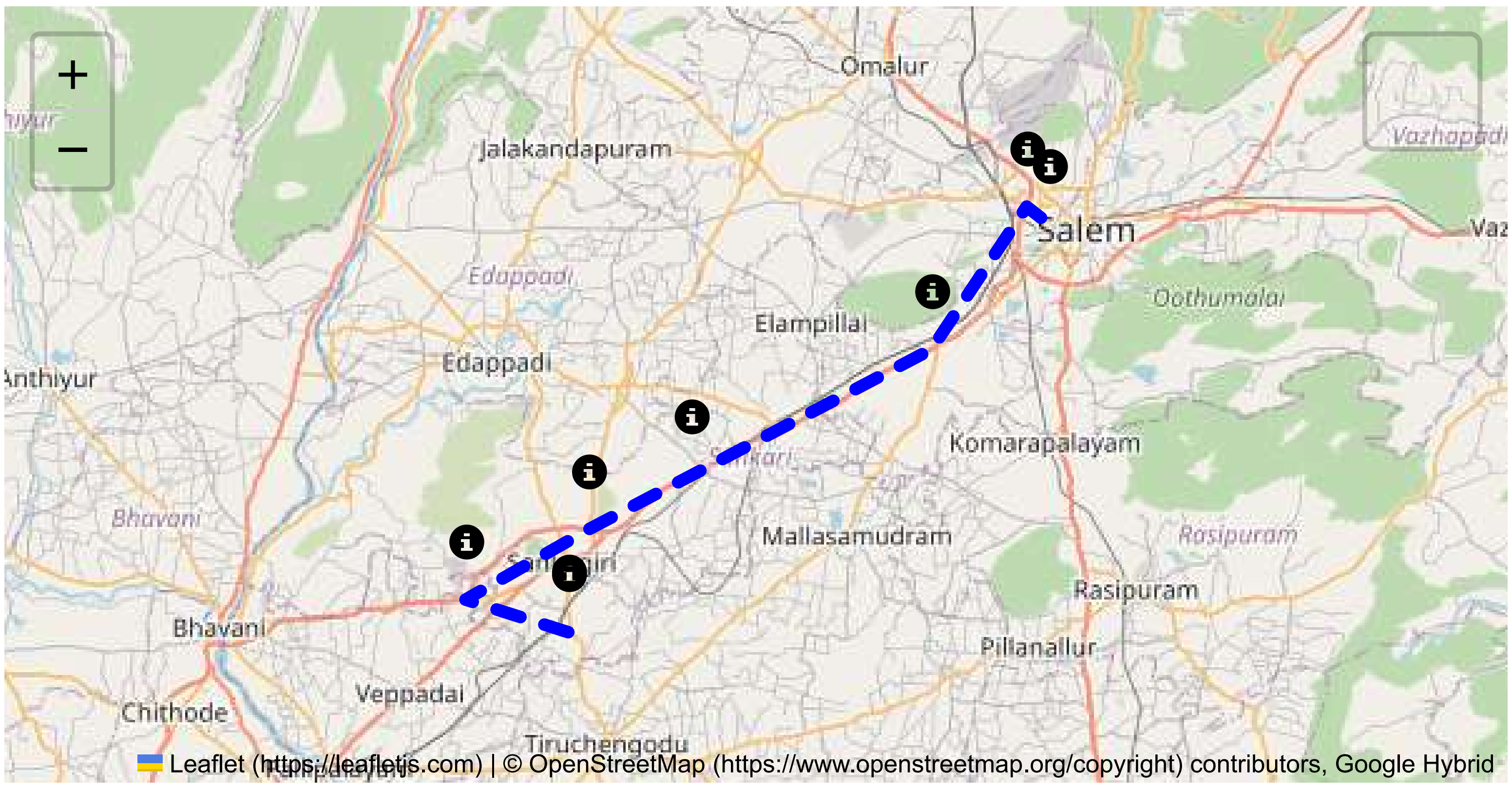
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

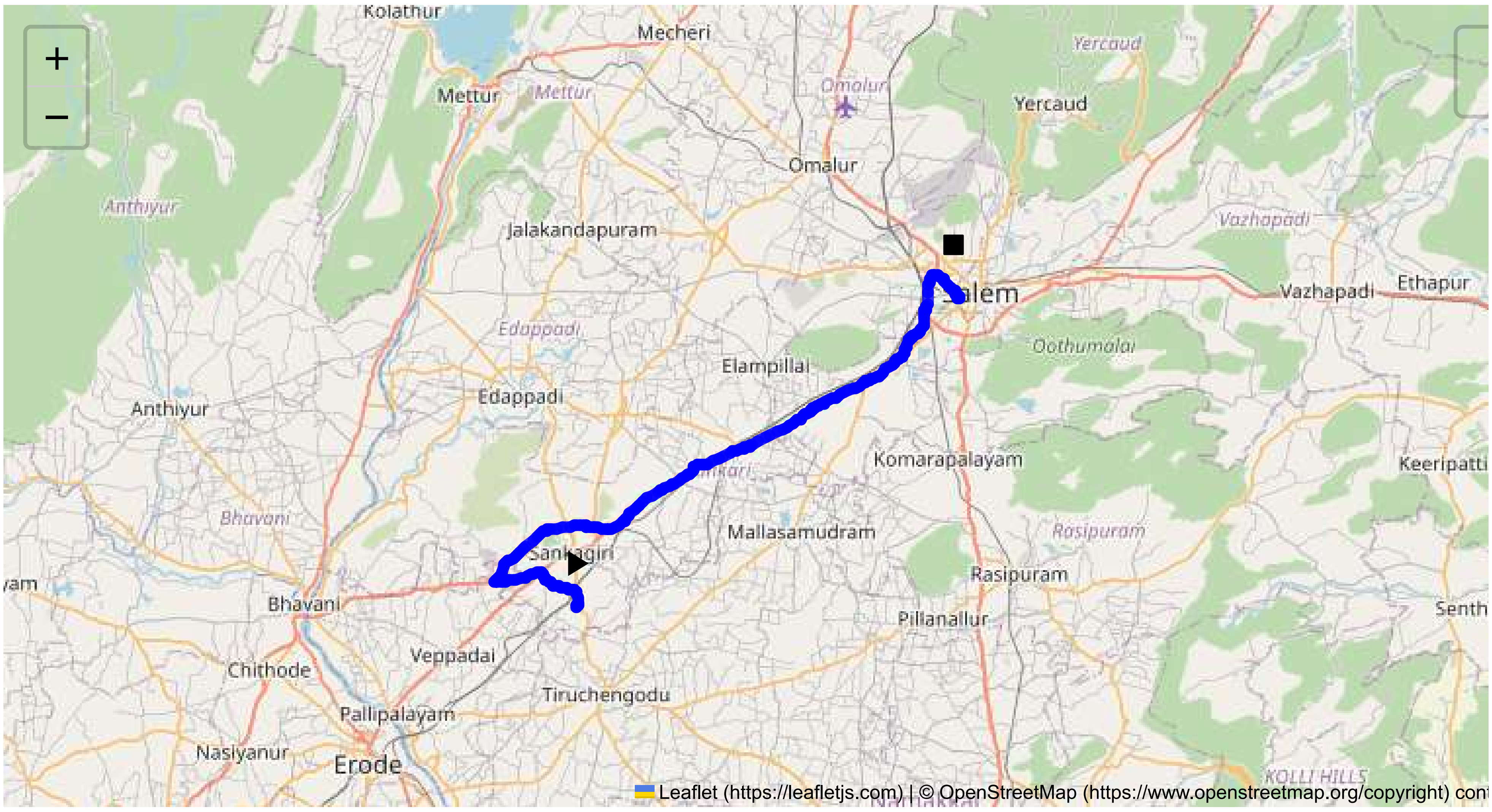
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 56.55 km
Estimated Duration: 1.2 hours
Adjusted Duration (Heavy Vehicle): 1.5 hours
Start: (11.4381, 77.8734)
End: (11.657941, 78.137024)



Welcome to the Journey Risk Management Study

Overview of the Route Map

The route runs from Sangagiri to Salem in Tamil Nadu, spanning approximately 56.55 kilometers. It navigates through several urban and semi-urban areas, including Puthur, Padaiveedu, and the Salem Bypass. The journey involves driving along Bhavani Main Road and the Salem-Kochi Highway, passing through areas like Vaikundam.

Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu generally experiences hot and humid weather. The route may encounter heavy rains during the monsoon (June to September), which could lead to water-logging and slippery roads, especially in the low-lying areas. Fog is rare but possible in the early mornings during the winter months (December to February).

Analysis of Traffic Patterns

- **Peak Hours**: Expect increased traffic during morning rush hours (8:00 AM - 10:00 AM) and evening hours (5:00 PM - 8:00 PM), particularly around Sangagiri and Salem.
- **Congestion-Prone Areas**: The Salem Bypass and major intersections, particularly near the entrances and exits of the towns along the route, are prone to traffic jams.

Assessment of Road Quality and Infrastructure

- **Road Quality**: The highways are generally well-maintained, but side roads closer to smaller towns can have potholes and uneven surfaces.
- **Infrastructure**: Adequate signage, but street lighting may be limited in rural stretches. Ensure care when driving at night.

Suggestions for Alternative Routes for Emergencies

- If the main route is blocked, consider taking local roads through adjacent towns, but be aware of potentially poor road conditions and lack of infrastructure.
- National Highway 44 can serve as an alternative pathway, connecting several towns parallelly to the route.

Summary of Local Regulations Affecting Hazardous Material Transport

- Transport of hazardous materials must comply with regulations on routing, which may restrict certain materials in proximity to populated areas.
- Night transportation might be advisable to avoid congestion but will need adequate lighting and marking of the vehicle.

Overview of Historical Incidents

Historical data indicates occasional accidents involving heavy vehicles in poor visibility conditions or during monsoons. However, no significant hazardous material incidents are noted, indicating general compliance with safety norms.

Environmental Considerations and Sensitive Areas

- **Sensitive Areas**: There are no major ecological reserves on this route, but rural areas may involve crossings close to agricultural lands.
- Avoid routes impacting local flora and fauna by sticking to designated highways.

Analysis of Communication Coverage

- Primary mobile network coverage is good along highways and major towns.
- Potential dead zones are likely in rural patches between larger towns, particularly between Padaiveedu and Vaikundam.

Estimated Emergency Response Times

- **Rural Segments**: Emergency response might take 30-45 minutes due to sparse medical facilities.
- **Urban Segments**: Within Salem, response time may drop to 10-20 minutes due to better infrastructure.

Overall Summary of Risk Assessment

This route is generally safe for transporting hazardous materials, provided weather conditions are monitored seasonally. Traffic congestion in urban areas can cause delays, but there are alternative routes for emergencies. Maintaining current knowledge of local regulations and ensuring adequate communication devices are crucial for risk management.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.66434, 78.13175	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43961, 77.87341	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	Medium	11.44898, 77.87410	30 KM/Hr
5	Turn	Medium	11.45357, 77.85789	30 KM/Hr
6	Turn	Medium	11.45352, 77.85698	30 KM/Hr
7	Turn	Medium	11.46048, 77.85036	30 KM/Hr
8	Turn	Medium	11.46303, 77.84945	30 KM/Hr
9	Turn	Medium	11.46318, 77.84913	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Turn	High	11.45631, 77.81668	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.49409, 77.88004	15 KM/Hr
13	Turn	Medium	11.49419, 77.88015	30 KM/Hr
14	Turn	Medium	11.49365, 77.88596	30 KM/Hr
15	Turn	High	11.49235, 77.89561	15 KM/Hr
16	Blind Spot	Blind Spot	11.49225, 77.89560	10 KM/Hr
17	Turn	Medium	11.64183, 78.11810	30 KM/Hr
18	Turn	High	11.66458, 78.12303	15 KM/Hr
19	Turn	High	11.66477, 78.12298	15 KM/Hr
20	Turn	High	11.66763, 78.12466	15 KM/Hr
21	Blind Spot	Blind Spot	11.65141, 78.14234	10 KM/Hr
22	Turn	Medium	11.65161, 78.14230	30 KM/Hr
23	Blind Spot	Blind Spot	11.65191, 78.14199	10 KM/Hr
24	Blind Spot	Blind Spot	11.65197, 78.14241	10 KM/Hr
25	Blind Spot	Blind Spot	11.65141, 78.14234	10 KM/Hr
26	Turn	Medium	11.65221, 78.14162	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
2	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium
3	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
4	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
5	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
6	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
9	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
13	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
14	hospital	Arokya Hospital, Pallapatty	11.6643647, 78.1301992	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
15	hospital	Ramalingam Hospital	11.6689149, 78.1311615	30 km/h	Medium
16	hospital	Sri Ragavendhra Neuro Care Pvt. Ltd.	11.6642728, 78.1321317	30 km/h	Medium
17	hospital	Neuro Foundation	11.664265, 78.13213	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
7	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
8	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
10	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
11	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
12	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium

Route Photos of Risky Spots

Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66434, 78.13175

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45357, 77.85789

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45352, 77.85698

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46048, 77.85036

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49409, 77.88004

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49365, 77.88596

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49235, 77.89561

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49225, 77.89560

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66458, 78.12303

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66477, 78.12298

Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66763, 78.12466

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.65141, 78.14234

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65161, 78.14230

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.65191, 78.14199

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.65197, 78.14241

Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.65141, 78.14234

Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65221, 78.14162